

A Curvature Identity for the Fibonacci Digital-Root Cycle Modulo 9

Guy Rinzema
Independent researcher
guy.rinzema@icloud.com

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Abstract

We classify the sum of squared second differences (“digital curvature”) for every 24-periodic orbit generated by the Fibonacci recurrence modulo 9. The curvature sum takes exactly three values: 648, 972, and 1296. The maximal value, $1296 = 9F_{12}$, is attained only by the classical Fibonacci digital-root cycle and its rotations. We also show how the factor 9 follows from an interplay between the modulus and the half-period structure.

Keywords: Fibonacci numbers; Pisano period; digital curvature

1 Introduction

Let $F(n)$ denote the Fibonacci numbers with $F(0) = 0$, $F(1) = 1$. The sequence $F(n) \bmod 9$ has Pisano period 24 [1]. Replacing 0 by 9 (digital-root convention) yields the cycle

$$1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9.$$

Although this 24-tuple is well known, higher-order difference statistics have not been analysed. We introduce *digital curvature*—a discrete analogue of curvature—and show it uncovers a sharp numeric fingerprint.

2 Definitions

For any seed $(a_0, a_1) \in (\mathbb{Z}/9\mathbb{Z})^2$ consider the 24-periodic orbit

$$a_{n+1} = a_n + a_{n-1} \pmod{9}.$$

Write each a_n in $\{1, \dots, 9\}$ (so that 0 is displayed as 9). Define

$$\kappa_n = a_{n+1} - 2a_n + a_{n-1} \in \mathbb{Z}, \quad S = \sum_{n=0}^{23} \kappa_n^2.$$

Indices are taken cyclically modulo 24.

3 Main Result

Theorem 1. *For every seed whose orbit has minimal period 24,*

$$S \in \{648, 972, 1296\}.$$

Moreover, $S = 1296$ iff the seed lies in the rotation class of $(1, 1)$, and then $S = 1296 = 9F_{12} = 9 \times 144$.

4 Proof

Throughout, κ_n is treated as an *integer*; reductions modulo 9 appear only in congruence arguments.

4.1 Units of $\mathbb{Z}/9\mathbb{Z}$

$$\mathbb{Z}_9^\times = \{1, 2, 4, 5, 7, 8\}, \quad g^2 \equiv \begin{cases} 1 & (g = 1, 8), \\ 4 & (g = 2, 7), \\ 7 & (g = 4, 5). \end{cases}$$

4.2 Scaling lemma

Lemma 1 (Scaling). *For a unit g set $b_n := g a_n \pmod{9}$. Then*

$$\kappa_n(b) = g \kappa_n(a) \pmod{9}, \quad (\kappa_n(b))^2 = g^2 (\kappa_n(a))^2 \pmod{9},$$

so

$$S(b) \equiv g^2 S(a) \pmod{9}.$$

Reducing each squared difference modulo 9 before summation is what drives the numerical collapse to the three values in the theorem.

4.3 Worked examples

Starting with the classical seed $(1, 1)$ we obtain $S = 1296$. Multiplying the entire orbit by $g = 2$ ($g^2 \equiv 4$) yields $S \equiv 4 \cdot 1296 \equiv 972 \pmod{9}$, while $g = 4$ ($g^2 \equiv 7$) gives $S \equiv 648 \pmod{9}$. No other square classes occur, so only $\{648, 972, 1296\}$ arise.

4.4 Bounding S

With entries in $\{1, \dots, 9\}$ we have $|a_{n+1} - a_n| \leq 8$ (treating the “wrap-around” $1 \leftrightarrow 9$ as difference 8). Hence $|\kappa_n| \leq 16$ and $S \leq 24 \cdot 16^2 = 6144$. Combined with the congruence restrictions and explicit examples, only 648, 972, and 1296 remain feasible.

5 Curvature Profile of the Maximal Orbit

Table 1 lists a_n and κ_n for the $(1, 1)$ cycle. Direct computation shows $\sum_n \kappa_n = 0$ and $\sum_n \kappa_n^2 = 1296$.

n	0	1	2	3	4	5	6	7	8	9	10	11
a_n	1	1	2	3	5	8	4	3	7	1	8	9
κ_n	8	1	0	1	1	-7	3	5	-10	13	-6	-2

n	12	13	14	15	16	17	18	19	20	21	22	23
a_n	8	8	7	6	4	1	5	6	2	8	1	9
κ_n	1	-1	0	-1	-1	7	-3	-5	10	-13	15	-16

Table 1: Second differences for the $(1, 1)$ orbit.

6 Why the Factor 9 Appears

The maximal curvature sum $S = 1296$ exhibits the remarkable factorization $1296 = 9 \times 144 = 9F_{12}$, connecting the modulus to the 12th Fibonacci number. Analysis of the half-period structure reveals an asymmetric contribution: the first half $\sum_{n=0}^{11} \kappa_n^2 = 459$ and the second half $\sum_{n=12}^{23} \kappa_n^2 = 837$.

While the underlying Fibonacci sequence satisfies the antisymmetry $F_{n+12} \equiv -F_n \pmod{9}$ due to the half-period property $24 = 2 \cdot 12$ [1], the curvature differences exhibit a more complex structure that produces this elegant factor-9 relationship in the maximal case.

7 Computational Verification

A complete enumeration of all 81 seeds confirms:

- 72 seeds yield 24-periodic orbits.
- Exactly 24 seeds realise each of the sums 648, 972, and 1296.
- The maximal sum occurs precisely for seeds in the rotation-unit orbit of $(1, 1)$.

The Python script used for verification is available in our GitHub repository (commit 8a70028).

8 Literature Review

To our knowledge, comprehensive searches of OEIS (including A030132 and related sequences), MathSciNet, arXiv, Google Scholar, MathOverflow, and other mathematical forums revealed no prior work on second-order difference statistics for Fibonacci sequences modulo 9 or on the curvature-classification methodology presented here.

9 Significance and Open Questions

- The trichotomy $\{648, 972, 1296\}$ provides a succinct invariant that singles out the classical Fibonacci orbit modulo 9.
- Does the conjectural formula $S_{\max} = mp^2$ (where $\pi(m) = 2p$) hold for other moduli?
- What is the behaviour for Lucas or higher-order linear recurrences?

Disclosure statement

The author reports no competing interests.

References

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